

1) Free electrons:

$$H_e = \sum_k \epsilon_k a_k^\dagger a_k, \quad \epsilon_k = \frac{k^2}{2m} \quad \left. \vphantom{\sum_k} \right\} \rightarrow H_{ei} = \sum_{\ell, \sigma} \int d\vec{r} \psi_\sigma^\dagger(\vec{r}) \underbrace{U(\vec{r} - \vec{R}_\ell)}_U \psi_\sigma(\vec{r})$$

$$H_p = \sum_{\vec{\epsilon}, \lambda} \omega_{\vec{\epsilon}, \lambda} \left(b_{\vec{\epsilon}, \lambda}^\dagger b_{\vec{\epsilon}, \lambda} + \frac{1}{2} \right)$$

$$N_{\vec{\epsilon}, \lambda} = (b_{\vec{\epsilon}, \lambda}^\dagger b_{\vec{\epsilon}, \lambda}) = \frac{1}{e^{\omega/T} - 1}$$

$$U(\vec{r} - \vec{R}_\ell) = \vec{\mu}_\ell \cdot \vec{\nabla} U(\vec{r} - \vec{R}_\ell)$$

$$H_{e-p} = - \sum_{\ell, \sigma} \int d\vec{r} \psi_\sigma^\dagger(\vec{r}) \left(\vec{\mu}_\ell \cdot \vec{\nabla} U(\vec{r} - \vec{R}_\ell) \right) \psi_\sigma(\vec{r})$$

$$\vec{\mu}_\ell = \frac{1}{N_i} \sum_{\vec{\epsilon}, \lambda} \sqrt{\frac{1}{2M\omega_{\vec{\epsilon}, \lambda}}} \vec{\epsilon}_{\vec{\epsilon}, \lambda} (b_{\vec{\epsilon}, \lambda} + b_{-\vec{\epsilon}, \lambda}^\dagger) e^{i\vec{\epsilon} \cdot \vec{R}_\ell}$$

$$\sum_{\ell} e^{i(\vec{k} - \vec{k}' + \vec{\epsilon}) \cdot \vec{R}_\ell} = N_i \sum_{\vec{G}} \delta_{\vec{k}, \vec{k}' + \vec{\epsilon} + \vec{G}}$$

$$U = -\frac{ze^2}{r} e^{-\Lambda r}$$

$$- \sum_{\ell, \sigma} \sum_{\vec{\epsilon}, \lambda} \frac{1}{\sqrt{2M\omega_{\vec{\epsilon}, \lambda} N_i}} \vec{\epsilon}_{\vec{\epsilon}, \lambda} \frac{c_{\vec{k}, \sigma}^\dagger c_{\vec{k}', \sigma}}{V} (b_{\vec{\epsilon}, \lambda} + b_{-\vec{\epsilon}, \lambda}^\dagger) e^{i\vec{\epsilon} \cdot \vec{R}_\ell} \int d\vec{r} e^{i(\vec{k} - \vec{k}') \cdot \vec{r}} \underbrace{\vec{\nabla} \left(\frac{1}{|\vec{r} - \vec{R}_\ell|} e^{-\Lambda |\vec{r} - \vec{R}_\ell|} \right)}_{\vec{\nabla} U(\vec{x})}$$

$\vec{r} - \vec{R}_\ell = \vec{x}$
 $\Lambda \vec{r} = i\vec{x}$

$$e^{i(\vec{k} + \vec{k} - \vec{k}') \cdot \vec{R}_\ell} \int e^{i(\vec{k} - \vec{k}') \cdot \vec{x}} \vec{\nabla} U(\vec{x})$$

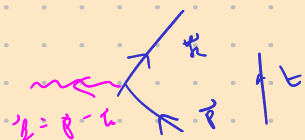
$$\sum_{\sigma, k, \lambda} \frac{i [\vec{\epsilon}_{\vec{\epsilon}, \lambda} \cdot (\vec{k} - \vec{k}')] }{\sqrt{2M\omega_{\vec{\epsilon}, \lambda}}} \frac{1}{V} \sqrt{N_i} U(\vec{k} - \vec{k}') \sum_{\vec{G}} \delta_{\vec{k}, \vec{k}' + \vec{\epsilon} + \vec{G}} c_{\vec{k}, \sigma}^\dagger c_{\vec{k}', \sigma} (b_{\vec{\epsilon}, \lambda} + b_{-\vec{\epsilon}, \lambda}^\dagger) =$$

$(\vec{G}=0)$ N: unklapp

$$= \frac{1}{\sqrt{V}} \sum_{\vec{k}, \lambda, \sigma} g_\lambda(\vec{k}, \vec{q}) c_{\vec{k} + \vec{q}, \sigma}^\dagger c_{\vec{k}, \sigma} (b_{\vec{\epsilon}, \lambda} + b_{-\vec{\epsilon}, \lambda}^\dagger) ; \quad g_\lambda(\vec{k}, \vec{q}) = -i \frac{1}{\sqrt{2M\omega_{\vec{\epsilon}, \lambda}}} \frac{4\pi ze^2}{(|\vec{\epsilon}|^2 + \Lambda^2)} \vec{\epsilon} \cdot \vec{\epsilon}_{\vec{\epsilon}, \lambda}$$

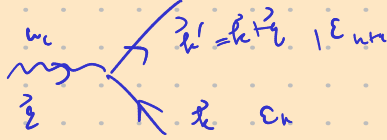
→ Fermi: spin ne drinja! $|\langle f | c_{\vec{k}}^\dagger c_{\vec{r}} | i \rangle|^2 = n_{\vec{k}} (1 - n_{\vec{k}})$

poln prazen

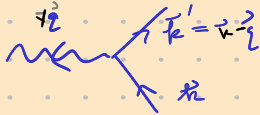


→ from trace: $K N_z - 1 |b_z| N_z \rangle|^2 = N_z$
 $| \langle N_z | b_z^\dagger | N_z + 1 \rangle |^2 = N_z + 1$

Fermi (variance): $\gamma_{i \rightarrow f} = 2\pi |K f | H_{fi} | i \rangle|^2 \underbrace{p(E_f)}_{\delta(E_f - E_i)}$

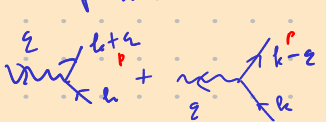
(1) Absorption process: 

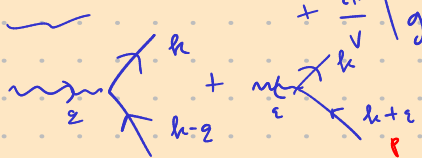
$|i\rangle = |N_z\rangle \otimes |k\rangle$
 $|f\rangle = |N_z+1\rangle \otimes |k+\vec{z}\rangle$
 $\rightarrow \sum_{k_z} \underbrace{\langle N_z+1 | (b_z + b_{-z}^\dagger) | N_z \rangle}_{N_z} \underbrace{\langle k+\vec{z} | c_{k+\vec{z}}^\dagger c_k | k \rangle}_{\sqrt{n_k (1-n_{k+\vec{z}})}} \cdot g \frac{1}{\sqrt{V}}$

(2) emission process: 

$|i\rangle = |N_z\rangle |k\rangle$
 $|f\rangle = |N_z+1\rangle |k-\vec{z}\rangle$
 $\rightarrow \sum_{k_z} \underbrace{\langle N_z+1 | b_z^\dagger | N_z \rangle}_{\sqrt{N_z+1}} \underbrace{\langle k-\vec{z} | c_{k-\vec{z}}^\dagger c_k | k \rangle}_{\sqrt{n_k (1-n_{k-\vec{z}})}} \cdot g \frac{1}{\sqrt{V}}$

(Mixture: $c_{k+\vec{z}}^\dagger c_k b_z + c_{k-\vec{z}}^\dagger c_k b_z^\dagger$)

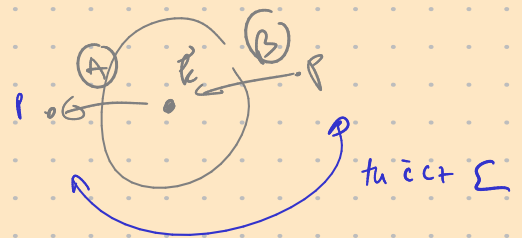
(A) $k \rightarrow p$: $\gamma_{k \rightarrow p} = \gamma_{abs} + \gamma_{em} = \frac{2\pi}{V} |g_\lambda(\vec{k}, \vec{z})|^2 n_k (1-n_{k+\vec{z}}) N_z \delta(\epsilon_{k+\vec{z}} - \epsilon_k - \omega_z) +$
 $+ \frac{2\pi}{V} |g_\lambda(\vec{k}, \vec{z})|^2 n_k (1-n_{k-\vec{z}}) (N_z+1) \delta(\epsilon_{k-\vec{z}} + \omega_z - \epsilon_k)$


(B) $p \rightarrow k$: $\gamma_{p \rightarrow k} = \gamma_{abs} + \gamma_{em} = \frac{2\pi}{V} |g_\lambda(\vec{k}, \vec{z})|^2 n_{k+\vec{z}} (1-n_k) (N_z+1) \delta(\epsilon_k + \omega_z - \epsilon_{k+\vec{z}}) +$
 $+ \frac{2\pi}{V} |g_\lambda(\vec{k}, \vec{z})|^2 n_{k-\vec{z}} (1-n_k) N_z \delta(\epsilon_k - \epsilon_{k-\vec{z}} - \omega_z)$


$N_{-z} = N_z$ then $\omega_z = \omega_{-z}$?

process inv. strength

$I_k = \sum_p \gamma_{p \rightarrow k} - \gamma_{k \rightarrow p} = \dots$



$= \frac{2\pi}{V} \sum_{\vec{z}, \lambda} |g_\lambda(\vec{k}, \vec{z})|^2 \left\{ (1-n_k) n_{k+\vec{z}} (1+N_z) \delta(\epsilon_k - \epsilon_{k+\vec{z}} + \omega_z) + \right.$
 $+ (1-n_k) n_{k-\vec{z}} N_z \delta(\epsilon_k - \epsilon_{k-\vec{z}} - \omega_z) -$
 $- n_k (1-n_{k-\vec{z}}) (1+N_z) \delta(\epsilon_{k-\vec{z}} - \epsilon_k + \omega_z) -$
 $\left. - n_k (1-n_{k+\vec{z}}) N_z \delta(\epsilon_{k+\vec{z}} - \epsilon_k - \omega_z) \right\}$

① $q \rightarrow -q$ $\hat{n}_k \rightarrow \hat{n}_{-k}$ $\hat{n}_{k'} \rightarrow \hat{n}_{-k'}$ $\delta(x-y) = \delta(y-x)$

$$\dot{n}_k = \frac{2\pi}{V} \sum_{k'} |g_\lambda(k, k')|^2 \left\{ (1-n_k) n_{k'} (1+N_2) \delta(\epsilon_k - \epsilon_{k'} + \omega_2) + \right. \\ \left. + (1-n_k) n_{k'} N_2 \delta(\epsilon_k - \epsilon_{k'} - \omega_2) - \right. \\ \left. - n_k (1-n_{k'}) (1+N_2) \delta(\epsilon_k + \epsilon_{k'} - \omega_2) - \right. \\ \left. - n_k (1-n_{k'}) N_2 \delta(\epsilon_k + \epsilon_{k'} + \omega_2) \right\} =$$

$$= \frac{2\pi}{V} \sum_{k'} |g_\lambda(k, k')|^2 \left\{ \delta(\epsilon_k - \epsilon_{k'} - \omega_2) \left[(1-n_k) n_{k'} N_2 - (1-n_{k'}) n_k (1+N_2) \right] + \right. \\ \left. + \delta(\epsilon_k - \epsilon_{k'} + \omega_2) \left[(1-n_k) n_{k'} (1+N_2) - n_k (1-n_{k'}) N_2 \right] \right\} =$$

Bei denselben:

✓ analog hier (1.23b)

$$\dot{n}_k = \sum_{k'} \Gamma_{k' \rightarrow k}^{\lambda} (1-n_k) n_{k'} - \Gamma_{k \rightarrow k'}^{\lambda} (1-n_{k'}) n_k$$

$$\Gamma_{k' \rightarrow k}^{\lambda} = \frac{2\pi}{V} |g_\lambda(k, k')|^2 \left\{ (1+N_2) \delta(\epsilon_k - \epsilon_{k'} + \omega_2) + N_2 \delta(\epsilon_k - \epsilon_{k'} - \omega_2) \right\}$$

$$\Gamma_{k \rightarrow k'}^{\lambda} = \frac{2\pi}{V} |g_\lambda(k, k')|^2 \left\{ (1+N_2) \delta(\epsilon_{k'} - \epsilon_k + \omega_2) + N_2 \delta(\epsilon_{k'} - \epsilon_k - \omega_2) \right\}$$

2. način:

$$n_k = \text{Tr}[\rho_e \cdot c_k^\dagger c_k]$$

$$N_z = \text{Tr}[\rho_e \cdot b_z^\dagger b_z]$$

$$\rho_0 = \rho_e \otimes \rho_h$$

$$H_{eq} = \frac{1}{\sqrt{V}} \sum_{k,q,\lambda} g_\lambda(k,q) c_{k+q}^\dagger c_k (b_{2\lambda} + b_{-2\lambda}^\dagger) \quad ; \quad \rho \equiv k_1 \rho$$

$$c_k(t) = e^{-i\epsilon_k t} c_k \rightarrow c_k^\dagger c_k(t) = e^{i(\epsilon_k - \epsilon_k)t} c_k^\dagger c_k$$

$$b_{2\lambda}(t) = e^{-i\omega_{2\lambda} t} b_{2\lambda}$$

$$\rightarrow H_{eq}^I(t) = \frac{1}{\sqrt{V}} \sum_{k,q,\lambda} g_\lambda(k,q) c_{k+q}^\dagger c_k (b_{2\lambda} e^{i(\epsilon_k - \epsilon_{k+q} - \omega_{2\lambda})t} + b_{-2\lambda}^\dagger e^{i(\epsilon_k - \epsilon_{k+q} + \omega_{2\lambda})t})$$

$$\rightarrow I_k^I(t) = \dot{n}_k^I(t) = i [H_{eq}^I(t), n_k] \quad \text{in} \quad \text{Tr}[\rho_0 I_k^I] = 0, \quad \text{in} \quad I \propto \langle b_{2\lambda} + b_{-2\lambda}^\dagger \rangle = 0$$

- $\mathcal{O}(\omega^2)$:

$$\dot{n}_k = i^2 \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T dt \text{Tr}(\rho_0 [H_{eq}^I(t), [H_{eq}, n_k]])$$

$[c_k, n_k] = \delta_{kk} c_k$
 $[c_k^\dagger, n_k] = -\delta_{kk} c_k^\dagger$

$$(1) \quad \frac{1}{\sqrt{V}} \sum_{k,q,\lambda} g_\lambda(k,q) (b_{2\lambda} + b_{-2\lambda}^\dagger) \underbrace{[c_{k+q}^\dagger c_k, n_k]}_{c_{k+q}^\dagger c_k \delta_{kk} - c_k^\dagger c_{k+q} \delta_{k+q,k}} =$$

$$= \frac{1}{\sqrt{V}} \sum_{q,\lambda} g_\lambda^{(1)}(q) (b_{2\lambda} + b_{-2\lambda}^\dagger) (c_{k+q}^\dagger c_k - c_k^\dagger c_{k+q})$$

$$(2) \quad \frac{1}{V} \sum_{k',q',\lambda'} \sum_{k,\lambda} g_{\lambda'}(k',q') g_\lambda(k,q) \left\{ (b_{2\lambda'}(t) + b_{-2\lambda'}^\dagger(t)) (b_{2\lambda} + b_{-2\lambda}^\dagger) [c_{k'+q'}^\dagger c_{k'}, c_{k+q}^\dagger c_k - c_k^\dagger c_{k+q}] + \right.$$

$$\left. + [b_{2\lambda'}(t) + b_{-2\lambda'}^\dagger(t), b_{2\lambda} + b_{-2\lambda}^\dagger] \cdot c_{k'+q'}^\dagger c_{k'} (c_{k+q}^\dagger c_k - c_k^\dagger c_{k+q}) \right\}$$

... born kamije, pravi deke.

2. BCS currents za ph-el. coupling:

$$k = k_{\uparrow} \uparrow \quad -k = k_{\downarrow} \downarrow$$

$$H_e = \sum_k \xi_k (c_k^\dagger c_k + c_{-k}^\dagger c_{-k}) - \sum_k \left(\Delta c_k^\dagger c_{-k}^\dagger + \Delta^* c_{-k} c_k \right) \quad \begin{matrix} \xi_k = \epsilon_k - \mu \\ \xi_{-k} = \xi_k \end{matrix}$$

$$\rightarrow E_k = \sqrt{\xi_k^2 + \Delta^2}, \quad M_k = \frac{1}{2} \left(1 + \frac{\xi_k}{E_k} \right), \quad N_k = \frac{1}{2} \left(1 - \frac{\xi_k}{E_k} \right), \quad M_k N_k = \frac{\Delta}{2E_k}$$

$$\rightarrow \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} M_k & N_k \\ -N_k & M_k \end{pmatrix} \begin{pmatrix} \gamma_k \\ \gamma_{-k}^\dagger \end{pmatrix} \Rightarrow H_e = E_0 + \sum_{k\sigma} E_k \gamma_{k\sigma}^\dagger \gamma_{k\sigma} \quad N_{k\sigma} = \langle \gamma_{k\sigma}^\dagger \gamma_{k\sigma} \rangle$$

→ enaka interakcija: $H_{ep} = \frac{1}{V} \sum_{\vec{k}, \vec{k}', \sigma} g_{\lambda}(\vec{k}, \vec{k}') c_{\vec{k}+\vec{k}', \sigma}^\dagger c_{\vec{k}, \sigma} (b_{\vec{k}'}^\dagger + b_{-\vec{k}'}^\dagger)$

rotiramo

rotiramo s novo bazo: $(M_k' = M_{k+2}, N_k' = N_{k+2}) \quad (M_k' = M_{-k}, N_k' = N_{-k}!)$

$$\begin{aligned} c_{k\uparrow} &= M_k \gamma_{k\uparrow} + N_k \gamma_{-k\downarrow}^\dagger \\ c_{k\uparrow}^\dagger &= M_k \gamma_{k\uparrow}^\dagger + N_k \gamma_{-k\downarrow}^\dagger \end{aligned}$$

$$\begin{aligned} c_{k\downarrow} &= M_k \gamma_{k\downarrow} - N_k \gamma_{-k\uparrow}^\dagger \\ c_{k\downarrow}^\dagger &= M_k \gamma_{k\downarrow}^\dagger - N_k \gamma_{-k\uparrow}^\dagger \end{aligned}$$

$$\begin{aligned} \downarrow \\ \sum_{k\sigma} c_{k+\vec{q}, \sigma}^\dagger c_{k\sigma} &= \sum_k \left[(M_{k+2} \gamma_{k+2\uparrow}^\dagger + N_{k+2} \gamma_{-k-2\downarrow}^\dagger) (M_k \gamma_{k\uparrow} + N_k \gamma_{-k\downarrow}^\dagger) + (M_{k+2} \gamma_{k+2\downarrow}^\dagger - N_{k+2} \gamma_{-k-2\uparrow}^\dagger) (M_k \gamma_{k\downarrow} - N_k \gamma_{-k\uparrow}^\dagger) \right] \\ &= \sum_{k\sigma} M_{k+2} M_k (\gamma_{k+2\uparrow}^\dagger \gamma_{k\sigma}) + \sum_{k\sigma} N_{k+2} N_k (\gamma_{-k-2\downarrow}^\dagger \gamma_{-k\sigma}^\dagger) + \\ &\quad + \sum_k (M_{k+2} N_k \gamma_{k+2\uparrow}^\dagger \gamma_{-k\downarrow}^\dagger - N_{k+2} M_k \gamma_{k+2\downarrow}^\dagger \gamma_{-k\uparrow}^\dagger) \stackrel{h \rightarrow -q-h}{=} (M_{k+2} N_k + M_k N_{k+2}) \gamma_{k+2\uparrow}^\dagger \gamma_{-k\downarrow}^\dagger \\ &\quad + \sum_k (N_{k+2} M_k \gamma_{-k-2\downarrow}^\dagger \gamma_{k\uparrow} - M_{k+2} N_k \gamma_{-k-2\uparrow}^\dagger \gamma_{k\downarrow}) \stackrel{h \rightarrow -q-h}{=} (M_{k+2} N_k + M_k N_{k+2}) \gamma_{-k-2\downarrow}^\dagger \gamma_{k\uparrow} \end{aligned}$$

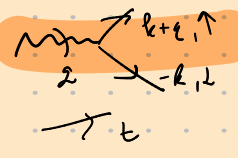
$$\begin{aligned} P_2^- &= \sum_{k\sigma} c_{k\sigma}^\dagger c_{k\sigma} = 2 \delta_{\vec{q}, 0} \sum_k N_k^2 + \sum_{k\sigma} (M_{k+2} M_k - N_{k+2} N_k) \gamma_{k+2, \sigma}^\dagger \gamma_{k, \sigma} + \\ &\quad + \sum_k (M_{k+2} N_k + N_{k+2} M_k) [\gamma_{k+2\uparrow}^\dagger \gamma_{-k\downarrow}^\dagger + \gamma_{-k-2\downarrow}^\dagger \gamma_{k\uparrow}] \end{aligned}$$

scattering
ce ali ne

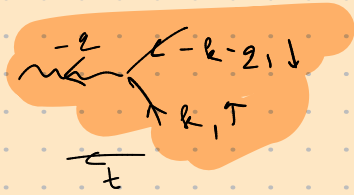
• scattering člena enaka, o faktorju: $|M_{k+2} M_k - N_{k+2} N_k|^2$

$$\hookrightarrow g_{\lambda}(\vec{k}, \vec{k}') \rightarrow G_{\lambda}^s(\vec{k}, \vec{k}') = g_{\lambda} (M_{k+2} M_k - N_{k+2} N_k)$$

• pair creation: $H_{eg}^c = \frac{1}{V} \sum_{k,q,\lambda} \left[\underbrace{G_{\lambda}^c(k,q)}_{G_{\lambda}^c(k,q)} (n_{k+q,\uparrow} n_{k,\downarrow} + n_{k+q,\downarrow} n_{k,\uparrow}) \right] \gamma_{k+q,\uparrow}^{\dagger} \gamma_{k,\downarrow}^{\dagger} (b_{2\lambda} + b_{-2\lambda}^{\dagger})$

$$|\langle f | H_{eg}^c | i \rangle|^2 \propto \underbrace{|N_{2-1} \langle b_{2\lambda} | N_{2-} \rangle|^2}_{N_2} \cdot (1 - n_{k+q,\uparrow}) (1 - n_{k,\downarrow})$$


• pair annihilation: $H_{eg}^a \propto \gamma_{k+q,\downarrow} \gamma_{k,\uparrow} (b_{2\lambda} + b_{-2\lambda}^{\dagger})$



$$|\langle f | H_{eg}^a | i \rangle|^2 \propto \underbrace{|N_{2-1} \langle b_{2\lambda}^{\dagger} | N_{2-} \rangle|^2}_{(N_{2-}+1)} \cdot n_{k,\uparrow} n_{-k-q,\downarrow}$$

calculate for:

$$p = k + q$$

$$\dot{n}_k = \sum_p \Gamma_{p \rightarrow k}^s (1 - n_k) n_p - \Gamma_{k \rightarrow p}^s n_k (1 - n_p) + \Gamma_{k,p}^c (1 - n_{k+q,\uparrow}) (1 - n_{k,\downarrow}) - \Gamma_{k,p}^a n_{k,\uparrow} n_{-k-q,\downarrow}$$

$$\Gamma_{p \rightarrow k}^s = \frac{2\pi}{V} \sum_{\lambda} |G_{\lambda}^s(k,p)|^2 \cdot \left\{ (1 + N_2) \delta(E_k - E_p + \omega_2) + N_{2-} \delta(E_k - E_p - \omega_2) \right\}$$

$$\Gamma_{k \rightarrow p}^s = \frac{2\pi}{V} \sum_{\lambda} |G_{\lambda}^s(k,p)|^2 \cdot \left\{ (1 + N_{2-}) \delta(E_p - E_k + \omega_2) + N_2 \delta(E_p - E_k - \omega_2) \right\}$$

$$\Gamma_{k,p}^c = \frac{2\pi}{V} \sum_{\lambda} |G_{\lambda}^c(k,p)|^2 \cdot \left\{ N_2 \delta(E_p + E_k - \omega_2) \right\}$$

$$\Gamma_{k,p}^a = \frac{2\pi}{V} \sum_{\lambda} |G_{\lambda}^a(k,p)|^2 \cdot \left\{ (1 + N_{2-}) \delta(\omega_2 - E_p - E_k) \right\}$$

→ v. standard major: $(u, v) \rightarrow (u, v)$ in (u, v) in (u, v)

major → $(u, v) \rightarrow (u, v)$ in (u, v) in (u, v)

(majority for pin up):

$$\begin{pmatrix} u & -v \\ v & u \end{pmatrix}, \text{ pin up}$$

$$\begin{pmatrix} u & v \\ -v & u \end{pmatrix}$$

→ key je: $A_{p \rightarrow k}$ (namer $\delta(E_p - E_k - \omega_2) \cdot \dots$???
 $\delta(p - k - \frac{1}{2})$?

3. EI currents, el. - ph.

$$H_e = \sum_k (c_{a,k}^\dagger, c_{b,k}^\dagger) \begin{pmatrix} \epsilon_a + v n_b & -\Delta^\dagger \\ -\Delta & \epsilon_b + v n_a \end{pmatrix} \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix}$$

$$\hookrightarrow \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} = \begin{pmatrix} m_k & r_k \\ -v_k & m_k \end{pmatrix} \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix}, \quad E_{\lambda k} = v_k \pm \sqrt{\xi_k^2 + \Delta^2};$$

$$\eta_k = \frac{\epsilon_a + \epsilon_b + v(n_a + n_b)}{2}, \quad \xi_k = \frac{\epsilon_a - \epsilon_b + v(n_a - n_b)}{2}$$

?

$$H_{ep} = \frac{1}{\sqrt{V}} \sum_{k, \lambda, \lambda'} (c_{a,k+\lambda}^\dagger, c_{b,k+\lambda}^\dagger) \begin{pmatrix} g_{a,\lambda}(k, \lambda) & 0 \\ 0 & g_{b,\lambda}(k, \lambda) \end{pmatrix} \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} (b_{\lambda,1}^\dagger + b_{-\lambda,1}^\dagger)$$

$$\begin{aligned} c_{a,k+\lambda}^\dagger c_{a,k} &= (u_p \alpha_k^\dagger + v_p \beta_k^\dagger) (u_k \alpha_k + v_k \beta_k) = \\ &= u_p u_k \alpha_k^\dagger \alpha_k + v_p v_k \beta_k^\dagger \beta_k + u_p v_k \alpha_k^\dagger \beta_k + v_p u_k \beta_k^\dagger \alpha_k \end{aligned}$$

$$H_{ep} = \frac{1}{\sqrt{V}} \sum_{k, \lambda, \lambda'} (\alpha_{k+\lambda}^\dagger, \beta_{k+\lambda}^\dagger) \begin{pmatrix} g_{a,\lambda}(k, \lambda) u_k u_{k+\lambda} + g_{b,\lambda}(k, \lambda) v_k v_{k+\lambda} & g_{a,\lambda}(k, \lambda) v_k v_{k+\lambda} - g_{b,\lambda}(k, \lambda) u_k u_{k+\lambda} \\ g_{a,\lambda}(k, \lambda) v_k v_{k+\lambda} - g_{b,\lambda}(k, \lambda) u_k u_{k+\lambda} & g_{a,\lambda}(k, \lambda) v_k v_{k+\lambda} + g_{b,\lambda}(k, \lambda) u_k u_{k+\lambda} \end{pmatrix} \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix} (b_{\lambda,1}^\dagger + b_{-\lambda,1}^\dagger)$$

$$G_{\mu\nu}(k, \lambda) = U_{\mu\lambda} (g_{\lambda\nu} \delta_{\lambda\nu}) U_{\nu\lambda} \quad \begin{matrix} \lambda, \nu = a, b \\ \mu, \lambda = a, b \end{matrix}$$

→ Fermi: $\mu, \nu = \{a, b\}$

$$\Gamma_{\mu k \rightarrow \nu k} = 2\pi \sum_i |\langle f | H_{ep} | i \rangle|^2 \delta(E_f - E_i)$$

$$G_{\mu\nu} \hat{c}_{a,k+\lambda}^\dagger \hat{c}_{b,k+\lambda} = \text{diagram} \quad \frac{1}{V} |G_{\mu\nu}|^2 n_{a,k} (1 - n_{a,k+\lambda}) N_{\lambda,2}$$

$$G_{\mu\nu} \hat{c}_{a,k+\lambda}^\dagger \hat{c}_{b,k+\lambda} = \text{diagram} \quad \frac{1}{V} |G_{\mu\nu}|^2 n_{a,k} (1 - n_{a,k+\lambda}) \cdot (1 + N_{\lambda,2})$$

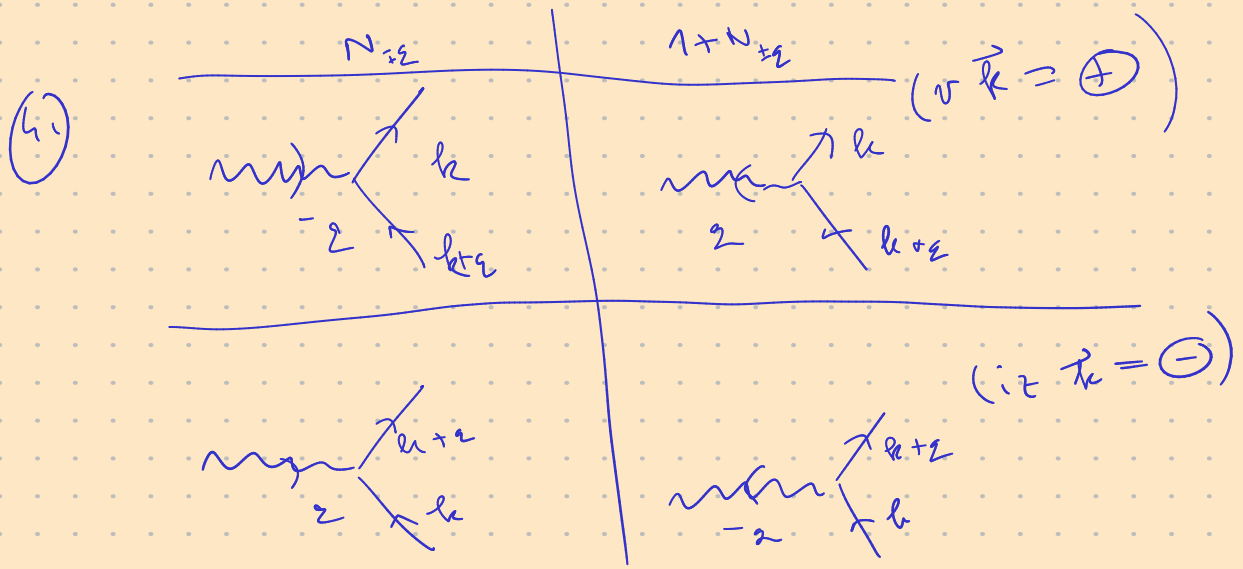
$\Gamma_{\nu p \rightarrow \mu k} = \text{diagram}$

$k \rightarrow (k) \xrightarrow{p}$

$$\dot{n}_{ak} = \sum_p \left(\Gamma_{ap \rightarrow ak} n_{ap} (1 - n_{ak}) - \Gamma_{ak \rightarrow ap} n_{ak} (1 - n_{ap}) + \Gamma_{bp \rightarrow ak} n_{bp} (1 - n_{ak}) - \Gamma_{ak \rightarrow bp} n_{ak} (1 - n_{bp}) \right)$$

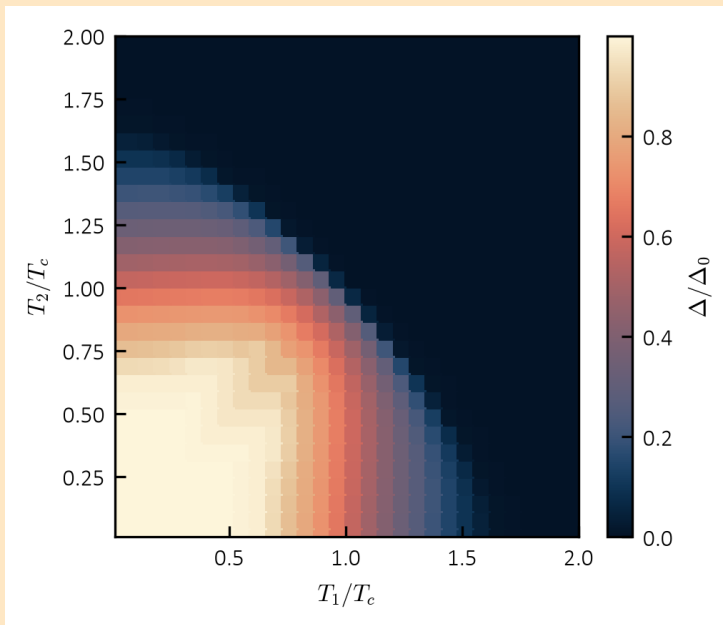
$$\Gamma_{\mu p \rightarrow \nu k} = \frac{2\pi}{V} \sum_{\lambda} |G_{\mu\nu}^\lambda(k, \lambda)|^2 \left\{ (1 + N_{\lambda,2}) \delta(E_{\nu k} - E_{\mu p} + \omega_{\lambda,2}) + N_{\lambda,2} \delta(E_{\nu k} - E_{\mu p} - \omega_{\lambda,2}) \right\}$$

$$\Gamma_{\mu k \rightarrow \nu p} = \frac{2\pi}{V} \sum_{\lambda} |G_{\mu\nu}^\lambda(k, \lambda)|^2 \left\{ (1 + N_{\lambda,2}) \delta(E_{\mu p} - E_{\nu k} + \omega_{\lambda,2}) + N_{\lambda,2} \delta(E_{\mu p} - E_{\nu k} - \omega_{\lambda,2}) \right\}$$



$$G^{\Lambda}(\vec{k}, \vec{z}) = \begin{pmatrix} \mu_{k+z} & \nu_{k+z} \\ -\nu_{k+z} & \mu_{k+z} \end{pmatrix} \begin{pmatrix} g_{\Lambda}(\vec{z}) & 0 \\ 0 & g_{\Lambda}(\vec{z}) \end{pmatrix} \begin{pmatrix} \mu_k & \nu_k \\ -\nu_k & \mu_k \end{pmatrix}$$

$$g_{\Lambda}(\vec{z}) = -i \frac{\hbar \omega z e^2}{\sqrt{2M\omega_{\Lambda}}} \cdot \frac{1}{|\vec{z}|^2 + \Lambda^2} (\vec{z} \cdot \vec{e}_{2\Lambda}^*)$$



4. free el. - photon int.

($c = -e$)

$$H = \frac{1}{2m} (\vec{p} + e\vec{A})^2 = \frac{\vec{p}^2}{2m} + \underbrace{\frac{e}{2m} \vec{p} \cdot \vec{A}}_{\substack{\text{"} \\ -i(\vec{p} \cdot \vec{A}) - i\vec{A} \cdot \vec{p} = \vec{A} \cdot \vec{p} \\ 0}} + \frac{e}{2m} \vec{A} \cdot \vec{p} + \frac{e^2}{2m} A^2 = \frac{\vec{p}^2}{2m} + \frac{e}{m} (\vec{A} \cdot \vec{p}) + \underbrace{\frac{e^2}{2m} A^2}_{\approx 0} \quad \mathcal{O}(A^2)$$

$$H = \sum_{\vec{k}\sigma} \epsilon_{\vec{k}} c_{\vec{k}\sigma}^\dagger c_{\vec{k}\sigma} + \frac{e}{m} \sum_{\vec{k}} \int d\vec{r} \psi_\sigma^\dagger(\vec{r}) [\vec{A} \cdot \vec{p}] \psi_\sigma(\vec{r})$$

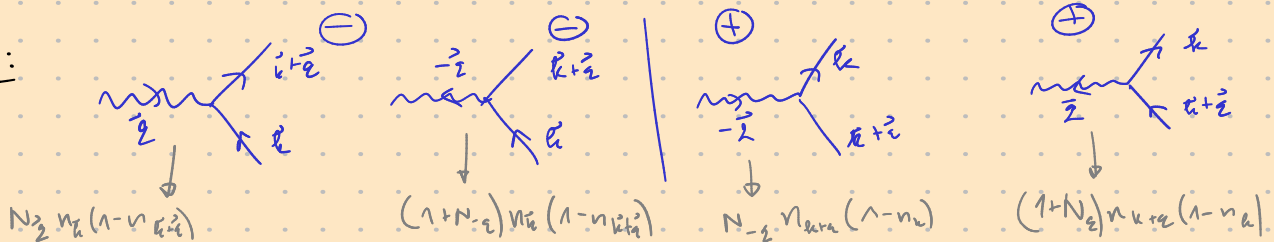
$A(\vec{r})$

(?)

$$\sum_{\vec{k}'} \sum_{\vec{k}''} \frac{e}{m} \int d\vec{r} e^{i(\vec{k}-\vec{k}')\vec{r}} c_{\vec{k}''\sigma}^\dagger \left[\frac{1}{\sqrt{2\epsilon_0 V \omega_{\vec{k}}}} \vec{\epsilon}_{\vec{k}} (a_{\vec{k}'} e^{i\vec{k}'\vec{r}} + a_{-\vec{k}'}^\dagger e^{i\vec{k}'\vec{r}}) \right] c_{\vec{k}\sigma} =$$

$$= \frac{1}{V} \sum_{\vec{k}, \vec{k}'} \frac{e}{\sqrt{2\epsilon_0 \omega_{\vec{k}}}} (\vec{\epsilon}_{\vec{k}} \cdot (\vec{k} + \vec{k}')) \cdot c_{\vec{k}+\vec{k}'}^\dagger c_{\vec{k}\sigma} (a_{\vec{k}} + a_{-\vec{k}}^\dagger) = 1e\gamma$$

→ Fermi:



$$\sum_{\vec{p}} \Gamma_{\vec{p} \rightarrow \vec{k}} = \sum_{\vec{p}, \vec{k}} \left(\frac{e^2}{2\epsilon_0 m^2 V \omega_{\vec{k}}} |\vec{\epsilon}_{\vec{k}} \cdot \vec{p}|^2 \right) \left\{ N_{-2} \delta(\epsilon_k - \epsilon_p - \omega_{\vec{k}}) + (1 + N_{-2}) \delta(\epsilon_k - \epsilon_p + \omega_{\vec{k}}) \right\}$$

$$\sum_{\vec{p}} T_{\vec{k} \rightarrow \vec{p}} = \sum_{\vec{p}, \vec{k}} \left(\frac{e^2}{2\epsilon_0 m^2 V \omega_{\vec{k}}} |\vec{\epsilon}_{\vec{k}} \cdot \vec{p}|^2 \right) \left\{ N_{-2} \delta(\epsilon_p - \epsilon_k - \omega_{\vec{k}}) + (1 + N_{-2}) \delta(\epsilon_p - \epsilon_k + \omega_{\vec{k}}) \right\}$$